

Myoglobin (1MBN)

Molecular Dynamics Simulation

Complete GROMACS Workflow
Including All Analysis and Output Files

Project By:
Krish Singh (@wasitkrish)

Project Reference:
<https://github.com/wasitkrish/Material-Informatics>

1. Install GROMACS

Download from <https://www.gromacs.org> or install on Ubuntu/Linux using:

```
sudo apt update  
sudo apt install gromacs
```

2. Clean PDB (Optional)

Remove unwanted HETATM records.

```
grep -v HETATM 1MBN.pdb > protein_clean.pdb
```

3. Generate Topology (processed.gro, topol.top, posre.itp)

Convert PDB into GROMACS format and generate topology files.

```
gmx pdb2gmx -f protein_clean.pdb -o processed.gro -water tip3p -ff amber99sb-ildn
```

4. Define Simulation Box (newbox.gro)

Create cubic box with 1.0 nm distance from protein.

```
gmx editconf -f processed.gro -o newbox.gro -c -d 1.0 -bt cubic
```

5. Solvate System (solv.gro)

Add water molecules to simulation box.

```
gmx solvate -cp newbox.gro -cs spc216.gro -o solv.gro -p topol.top
```

6. Add Ions (ions.tpr, solv_ions.gro)

Neutralize the system.

```
gmx grompp -f ions.mdp -c solv.gro -p topol.top -o ions.tpr
gmx genion -s ions.tpr -o solv_ions.gro -p topol.top -pname NA -nname CL -neutral
```

7. Energy Minimization (em.*)

Preprocess, run minimization and extract potential energy.

```
gmx grompp -f minim.mdp -c solv_ions.gro -p topol.top -o em.tpr
gmx mdrun -deffnm em
gmx energy -f em.edr -o potential.xvg
```

8. NVT Equilibration (nvt.*)

Equilibrate temperature.

```
gmx grompp -f nvt.mdp -c em.gro -r em.gro -p topol.top -o nvt.tpr
gmx mdrun -deffnm nvt
gmx energy -f nvt.edr -o temperature.xvg
```

9. NPT Equilibration (npt.*)

Equilibrate pressure and density.

```
gmx grompp -f npt.mdp -c nvt.gro -r nvt.gro -t nvt.cpt -p topol.top -o npt.tpr
gmx mdrun -deffnm npt
gmx energy -f npt.edr -o pressure.xvg
gmx energy -f npt.edr -o density.xvg
```

10. Production MD (md.*)

Run 5 ns production simulation.

```
gmx grompp -f md.mdp -c npt.gro -t npt.cpt -p topol.top -o md.tpr
gmx mdrun -deffnm md
```

11. Trajectory Processing (md_center.xtc, md_fit.xtc)

Remove PBC and fit trajectory.

```
gmx trjconv -s md.tpr -f md.xtc -o md_center.xtc -center -pbc mol
gmx trjconv -s md.tpr -f md_center.xtc -o md_fit.xtc -fit rot+trans
```

12. Complete Structural & Energy Analysis

Generate ALL analysis outputs including total energy.

```
gmx rms -s md.tpr -f md_fit.xtc -o rmsd.xvg
gmx rmsf -s md.tpr -f md_fit.xtc -o rmsf.xvg
gmx gyrate -s md.tpr -f md_fit.xtc -o gyrate.xvg
gmx hbond -s md.tpr -f md_fit.xtc -num hbnum.xvg
gmx sasa -s md.tpr -f md_fit.xtc -o sasa.xvg
gmx do_dssp -s md.tpr -f md_fit.xtc -o dssp.xpm -sc dssp.xvg
gmx xpm2ps -f dssp.xpm -o dssp.ps
gmx energy -f md.edr -o total_energy.xvg gmx energy -f md.edr -o potential_md.xvg
gmx energy -f md.edr -o temperature_md.xvg gmx energy -f md.edr -o pressure_md.xvg
gmx energy -f md.edr -o density_md.xvg
```

13. Extract Final Structure (final.pdb)

Extract structure at 5000 ps (5 ns).

```
gmx trjconv -s md.tpr -f md.xtc -o final.pdb -dump 5000
```
